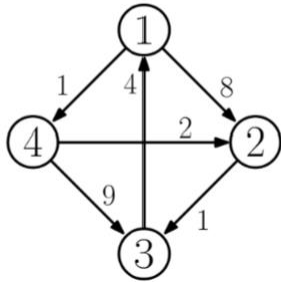


CS385Q Homework 5: Dynamic Programming

1. Apply the Floyd-Warshall Algorithm to find the all-pair shortest paths for the following graph. State the shortest paths for each vertex to every other vertex. (10 points)



2. Compute the transitive closure for the above graph. Show each intermediate step. (10 points)

3. Given the following coins, fill out the following table for finding the maximum amount of money subject to the constraint that no two adjacent coins can be picked up. State which coins were picked up to compute the maximum value. (10 points)

Index i	0	1	2	3	4	5	6
C_i	0	3	5	2	6	4	1
$F(i)$							
$S(i)$							

4. Given the following items and a knapsack with weight capacity 5, find the most valuable subset of the items that fits in the knapsack. Fill out the table for $F(i,j)$ and state the items in the knapsack. (10 points)

i	1	2	3	4
w_i	2	1	3	2
v_i	15	11	21	13

$i \setminus j$	0	1	2	3	4	5
0						
1						
2						
3						
4						